

Darshini Shah

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GitHub LinkedIn Leetcode Codeforces

EDUCATION

- **Indian Institute Of Technology Patna** Bihar, India
Bachelor of Technology - Artificial Intelligence & Data science July 2024 - May 2028
 - **CGPA: 8.72/10**
 - **SGPA(3rd sem): 9.64/10 (Among the top 5 performers in the branch)**
 - **Courses:** Data Structures and Algorithms, AI Concepts, Optimization Techniques, Digital Logic and Computer Organization, Discrete Mathematics , Database and Management System

PROJECTS

- **Jilo**
 - Built an **AI-powered RCM normalization engine** to process unstructured hospital documents using **OCR (pdf2image, EasyOCR)** and **Gemini-based extraction** for clinical and financial entities.
 - Designed a structured pipeline with **prompt engineering, JSON schema enforcement, and HL7 FHIR (R4) bundle generation** using a **FastAPI backend** and **React dashboard**.
- **Advanced RAG System**
 - Developed a **hybrid retrieval system** combining **Gemini embeddings + ChromaDB (vector search)** and **BM25 (keyword search)** to improve retrieval accuracy.
 - Implemented **query decomposition, parallel retrieval, Cohere reranking, and LLM-based evaluation** with **timestamped citation linking** for reliable responses.
- **Smart Allocator**
 - Developed an **AI-powered volunteer allocation system** using **geospatial filtering, skill matching, and availability constraints** for efficient resource distribution.
 - Integrated **Gemini-based OCR with MongoDB GridFS** and implemented **real-time notifications, leaderboard system, and automated role assignment** for scalable coordination.

SKILLS

- **Languages:** Python, C++, C, JavaScript
- **Core CS:** Data Structures & Algorithms, DBMS, OOPs
- **Backend & Systems:** Node.js, FastAPI, REST APIs, WebSockets, System Design Basics
- **Databases:** MongoDB, SQL, Supabase
- **Auth & Security:** JWT Authentication
- **AI & LLMs:** RAG, LangChain, LangGraph, Prompt Engineering, Text Processing
- **Machine Learning:** Supervised Learning, Deep Learning (CNNs, RNNs)
- **Tools & Frameworks:** Docker, Git, Scikit-learn
- **Data Handling:** Data Preprocessing, PDF Parsing, Feature Engineering

ACHIEVEMENTS

- Secured **2nd place** at **HACKMATRIX 2.0** (Healthcare Hackathon)
- Selected as a **core contributor** for an **AI-powered PR Reviewer Bot** (Open Source Program)
- Ranked **125,000** among **5M+ users** on **LeetCode** (Rating: **1446**; Solved **600+** DSA problems)
- **Codeforces Pupil** (Rating: **1200+**)
- **JEE Advanced AIR 9272**

EXPERIENCE

- **Design Team Coordinator, TEDxIITPatna** Patna, India
May 2025 – Present
 - **Led the design team** for TEDx event creatives, utilizing **Figma and Adobe Photoshop** for marketing and promotional materials across multiple platforms.
- **Inter IIT Tech Meet 14.0** PS by **ObservAI**
 - **Developed** an agentic query-processing chatbot enabling multi-turn conversational handling.
 - **Architected** an end-to-end pipeline with query decomposition and **optimized routing** using priority queues and tree-based strategies.